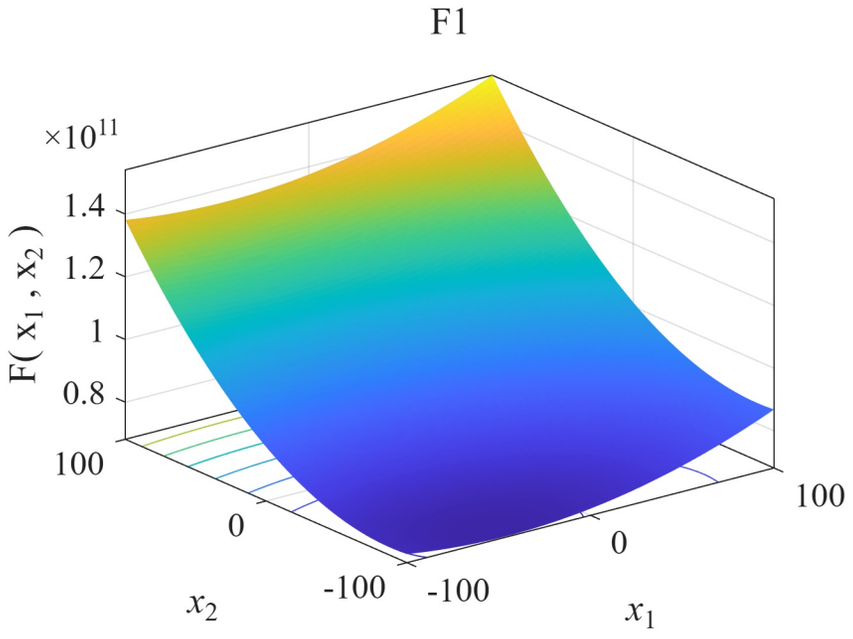
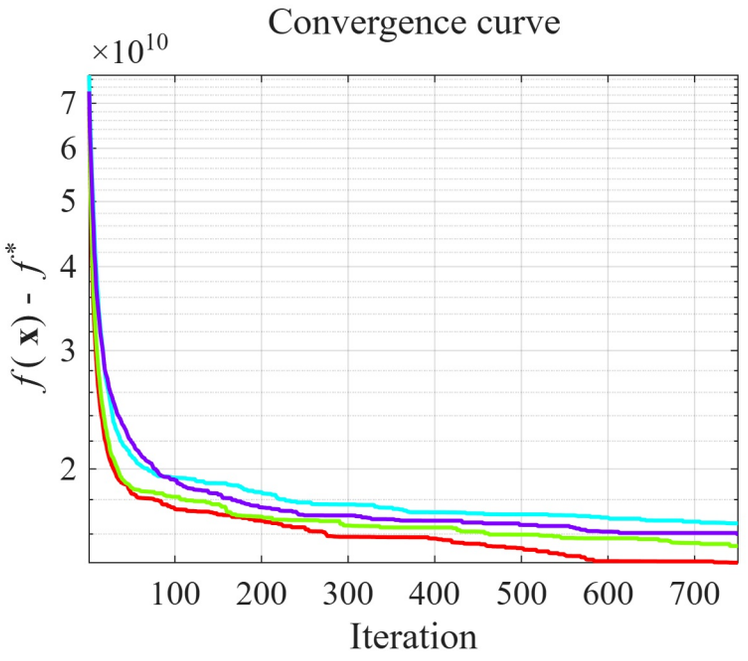
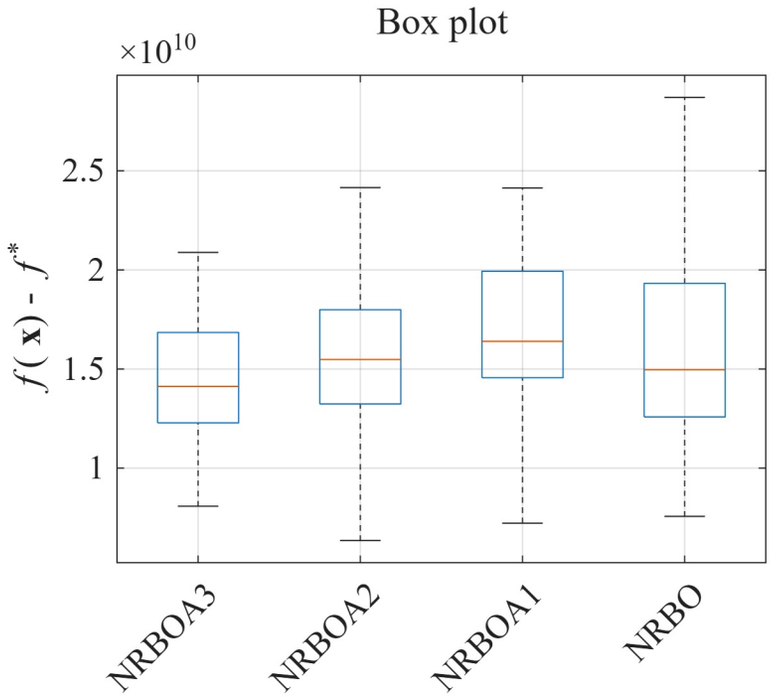

F1
×10^11
1.4
1.2
1
0.8
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1


Convergence curve
×10^10
7
6
5
4
3
2
f(x) - f*
100
200
300
400
500
600
700
Iteration


Box plot
×10^10
2.5
2
1.5
1
f(x) - f*
NRBOA3
NRBOA2
NRBOA1
NRBO

NRBOA3
NRBOA2
NRBOA1
NRBO